Syed Ahsan Masud Zaidi

Ph.D. Candidate in Computer Science | Kansas State University
Manhattan, Kansas, USA | ahsanzaidi@ksu.edu | +1 (785) 320-0756
Google Scholar | GitHub | LinkedIn | ORCID

Computer-vision researcher focused on sports motion analysis and athlete safety, with work in zero-shot event localization, video action classification, and public research datasets. University teaching experience at K-State, UET Lahore, and KFUPM.

Education

Ph.D., Computer Science 2023–Present

Kansas State University, USA. Candidate; GPA: 3.96/4.0. Advisor: Lior Shamir.

M.S., Electrical Engineering 2019

King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia.

Thesis research: laser-beam position control for free-space optical communication.

B.S., Mechatronics & Control Engineering 2013

University of Engineering & Technology (UET), Lahore, Pakistan.

Thesis: *Static Marker-Based Cricket Bowling Analysis*.

Publications

[1] **Zaidi, S. A. M.**, Hsu, W., and Dietrich, S. "ViTs for Action Classification in Videos: An Approach to Risky Tackle Detection in American Football Practice Videos." *International Conference on Pattern Recognition (ICPR 2026)*, LNCS 16826, pp. 605–620, Springer Nature, 2027. Published online August 3, 2026.

[2] Zaidi, T., Munir, A., and **Zaidi, S. A. M.** "CRISP: Context-Robust Skill Inference for Long-Horizon Offline Robotic Control." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. Accepted.

[3] **Zaidi, S. A. M.**, Shamir, L., and Dietrich, S. "Reproducing Vision Transformer-Based Risky Tackle Detection in American Football Practice Videos: A Companion Study." *Reproducible Research in Pattern Recognition (RRPR), Workshop at ICPR*, 2026. Accepted for Springer LNCS publication; **oral presentation**. Conference listing. Related preprint: "Revisiting Risky Tackle Detection with Vision Transformers," arXiv:2609.35562.

[4] **Zaidi, S. A. M.**, Shamir, L., Hsu, W., Dietrich, S., and Zaidi, T. "GRAZE: Grounded Refinement and Motion-Aware Zero-Shot Event Localization." *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVSports)*, pp. 10087–10095, 2026. **Spotlight**.

[5] Khan, R. A., Asad, A., Jamal, M. S., and **Zaidi, S. A.** "Olympus Field Development Optimization Study – KFUPM." *EAGE/TNO Workshop on OLYMPUS Field Development Optimization*, 2018.

Preprint

[6] Nayab, S., Chohan, S. R., Jameel, A., Shah, S. R., **Zaidi, S. A. M.**, Jha, A. N., and Siddique, K. "Advancing Remote and Continuous Cardiovascular Patient Monitoring through a Novel and Resource-efficient IoT-Driven Framework." 2025. arXiv:2505.03409.

Research Experience

Doctoral Research, Computer Vision and Video Understanding 2023–Present

Department of Computer Science, Kansas State University

Graduate Research Assistant

- Developed **GRAZE**, a zero-shot contact-event localization framework using grounded object detection, segmentation, and motion-aware temporal refinement, without task-specific model training. [4]
- Developed vision-transformer methods for risky-tackle classification on a 733-clip subset of TackleNet, with class-imbalance handling and cross-validation. [1]
- Built reproducible training and evaluation workflows with session identifiers and session-disjoint protocols; benchmarked ViViT, MViTv2, SlowFast, and VideoMAE v2.
- Collaborated with Lior Shamir, William Hsu, and athletic-training expert Scott Dietrich on video analysis of tackling technique and athlete safety.

Public Dataset and Software

TackleNet: American Football Practice Tackle Videos 2026

Released **737 de-identified clips from 30 practice sessions**, with six SATT component scores assigned by a domain expert and first-point-of-contact annotations. Dataset | Benchmark code.

Teaching Experience

Kansas State University 2023–Present

Graduate Teaching Assistant (regular semesters), Department of Computer Science

Course instructor, CIS 209: Computer Programming for Engineers Spring 2026 and Fall 2026

- Teach Python programming through guided coding exercises and labs; develop assessments, grade student work, and provide individual support that emphasizes problem decomposition and independent reasoning.
- Revised course materials using Canvas accessibility feedback and adjusted assessment weights to emphasize independently demonstrated understanding.
- **TA mentoring:** Provided guidance on programming instruction and student support in Spring 2026; continue this mentoring in Fall 2026.

Teaching support, CIS 209 Spring 2023–Fall 2025

Supported labs, office hours, debugging, and student feedback.

Grader, CIS 200: Programming Fundamentals Summer 2025

University of Engineering & Technology, Lahore 2014–2023

Lecturer, Department of Mechatronics & Control Engineering, Pakistan

- Taught undergraduate programming, embedded systems, digital logic, signals and systems, circuits and electronics, and engineering mechanics.
- Served as Director of the Simulation Lab and In-Charge of the Semester Cell.

King Fahd University of Petroleum & Minerals

Teaching Assistant / Lab Instructor, Dhahran, Saudi Arabia

Supported Neural Networks Theory & Applications (EE 555), Image Processing (EE 663), and Engineering Physics laboratories (PHYS 101/102).

Honors, Scholarships, and Recognition

- **Playmaker Grant Lab Finalist**, American Orthopaedic Society for Sports Medicine (AOSSM), 2026.
- **Department head nomination**, GSC Award for Graduate Student Teaching Excellence, Kansas State University, 2026.
- **Broadening Participation support**, Computer Vision Foundation, CVPR 2026.
- **Travel Stipend and Registration Scholarship**, International Association for Pattern Recognition (IAPR), ICPR 2026.
- **Department of Computer Science Travel Award**, Kansas State University, 2026; support for presenting at ICPR in Lyon, France.
- **Full Merit Scholarship**, M.S. studies, KFUPM.

Conference Presentations

- **Oral presentation**, RRPR Workshop at ICPR, Lyon, France, August 2026. Reproducibility of vision-transformer risky-tackle detection. [3]
- **Poster**, ICPR main conference, Lyon, France, August 2026. Vision transformers for risky-tackle classification. [1]
- **Research proposal pitch**, AOSSM Playmaker Grant Lab, Annual Meeting, Seattle, Washington, July 10, 2026. "Detection & Analysis of Risky Football Tackles Using Artificial Intelligence."
- **Spotlight and poster**, CVSports Workshop at CVPR, Denver, Colorado, June 2026. GRAZE: zero-shot event localization. [4]
- **Poster**, CVSports Workshop at CVPR, Denver, Colorado, June 2026. Vision transformers for risky-tackle classification. [1]
- **Oral presentation**, K-State Graduate Research, Arts, and Discovery (KGRAD) Forum, April 2026. Vision transformers for risky-tackle detection.
- **Poster**, K-State AI Symposium, October 2025. AI for athlete safety and injury prevention.

Academic Service

Peer reviewer

- Journals: *IEEE Access*; *SLAS Technology*.
- Workshops: AI4Science Workshop at ICML 2026; WiCV Workshop at ECCV 2026; NewInML Workshop at NeurIPS 2026; RRPR Workshop at ICPR 2026.

15 verified peer reviews, Web of Science (October 4, 2026).

Technical Expertise

Programming: Python, MATLAB, C, C++.

Machine learning and vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face Transformers; video transformers, Grounding DINO, SAM2.

Tools and computing: Git, Linux, LaTeX; GPU training on K-State Beocat and PSC Bridges-2.

Engineering: Signal processing, control systems, LabVIEW, embedded systems.

Earlier Professional Experience

Founder and CEO, Black Gold Solutions 2014

Technology startup at Plan9, Punjab Information Technology Board, Lahore. Third prize, TiE Startup Challenge 2014.

Project Engineer, RBL PepsiCo 2013–2014

Commissioned a KRONES PET filling plant; supported system integration and production handover.